FIITJEE

ALL INDIA TEST SERIES

JEE (Advanced)-2023

CONCEPT RECAPITULATION TEST – I

PAPER –1

TEST DATE: 26-04-2023

Time Allotted: 3 Hours **Maximum Marks: 180**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-B**.

Section – A *(01 –06, 19 – 24, 37 – 42):* This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).

Section – A *(07 – 10, 25 – 28, 43 – 46):* This section contains **TWELVE (12)** Matching List Type Questions. Each question has **FOUR** statements in **List-I** entries (I), (II), (III) and (IV) and **FIVE** statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which, **ONLY ONE** of these four options is correct answer.

Section – B *(11 – 18, 29 – 36, 47 – 54):* This section contains **TWENTY FOUR (24)** numerical based questions. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

MARKING SCHEME

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	if three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct numerical value is entered at the designated place;
Zero Marks	:	0	In all other cases.

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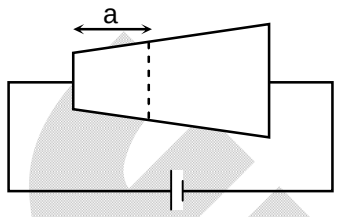
Physics

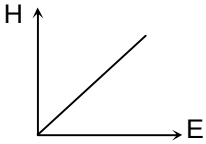
PART – I

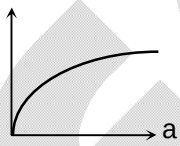
Section – A (Maximum Marks: 24)

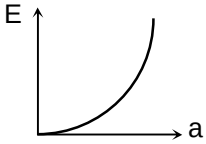
*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

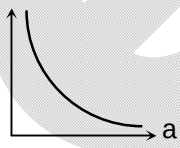
1. The electric field intensity, potential and rate of heat generation per unit length are E, V and H at a section distant "a" from left end of a conical shaped conductor connected to a battery of emf E, as shown. Find correct options:



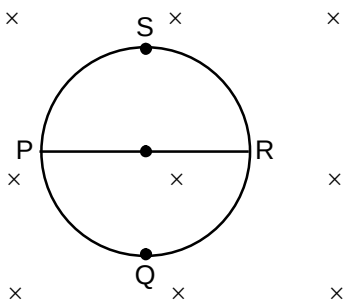
(A) 

(B) 

(C) 

(D) 

2. The radius of the circular loop is 'a'. Magnetic field is increasing at the constant rate is α. Magnetic field is confined to a cylindrical region and axis of magnetic field coincides with the axis of the loop. Resistance per unit length of the wire of loop is ρ. Choose the correct option (s)



(A) Current in the loop PQRS is $\frac{a\alpha}{2\rho}$ anticlockwise

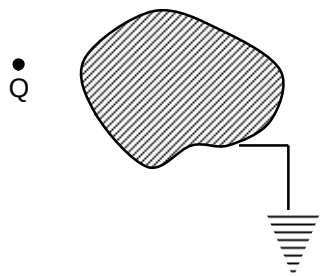
(B) Current in the loop PQRS is $\frac{a\alpha}{\rho}$ clockwise

(C) Current in the wire PR is zero

(D) Current in the wire PR is $\frac{\pi a\alpha}{2\rho}$

3. A point object is placed at 30 cm from a convex glass lens (μ_g = 3/2) of focal length 20 cm. The final image of object will be formed at infinity if
(A) Another concave lens of focal length 60 cm is placed in contact with the previous lens.
(B) The whole system is immersed in a liquid of refractive index 4/3
(C) Another convex lens of focal length 60 cm is placed at a distance of 30 cm from the first lens
(D) The whole system is immersed in a liquid of refractive index 9/8.

4. An arbitrary conductor (grounded) is kept at same distance from a point charge 'Q'. Which of the following options are correct?
(A) The magnitude of total charge on the surface of conductor will always be less than Q.
(B) The magnitude of total charge on the surface of conductor can be greater than or equal to Q. depending on its geometry.
(C) Net charge on the conductor with always be negative, but on some points on its surface, charge can be positive.
(D) Net charge on the conductor will always be negative and there can't be any point on its surface, where charge is positive.

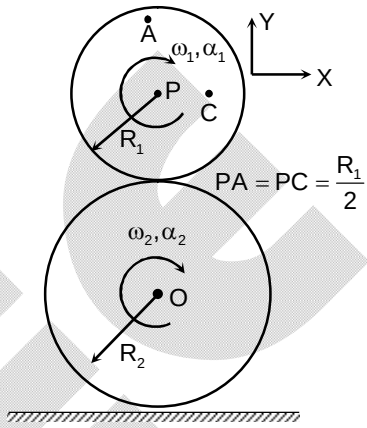


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5. In a chamber of volume 70 litre, a gas is filled at initial temperature of 77°C and pressure 4.155×10^5 N/m^2. The gas does 8077.5 Joule work by providing heat by a source of constant power 100 watt for 1 minute and the temperature of the gas decreases by 10°C. Given that gas constant R = 8.31 J/mole°K. Then

(A) Gas in the chamber is mono atomic
(B) Gas in the chamber is diatomic
(C) No. of moles of gas in the chamber is 8
(D) No. of moles of gas in the chamber is 10

6. Lower disc is fixed at its centre, and upper disc is doing pure rolling over it. Which of the following are correct?



(A) $\vec{a}_A = (2\alpha_1 R_1 + \alpha_2 R_2)\hat{i} - \left[\omega_1^2 R_1 + \frac{(\omega_1 R_1 + \omega_2 R_2)^2}{R_1 + R_2}\right]\hat{j}$

(B) KE of the upper disc is $\frac{m}{4}\left[3\omega_1^2 R_1^2 + 2\omega_2^2 R_2^2 + 4\omega_1\omega_2 R_1 R_2\right]$

(C) $\vec{L}$ upper disc about O
$= \left[\frac{3}{2}m\omega_1 R_1^2 + m\omega_2 R_2^2 + mR_1 R_2(\omega_1 + \omega_2)\right](-\hat{k})$

(D) $\vec{L}$ upper disc about A $\neq$ $\vec{L}$ disc about C

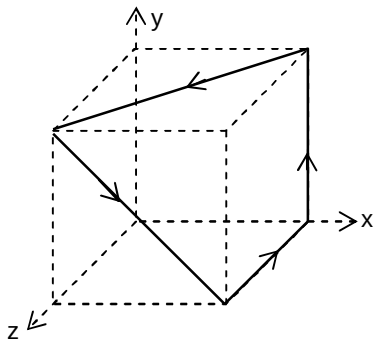
(Given mass of upper disc = m)

[ω_1, ω_2 : Angular speed of both discs respectively
α_1, α_2 : Angular acceleration of both discs respectively]

Section – A (Maximum Marks: 12)

*This section contains **FOUR (04)** Matching List Type Questions. Each question has FOUR statements in **List-I** entries (I), (II), (III) and (IV) and FIVE statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.*

7. A wire carrying a 10A current is bent to pass through various sides of a cube of side 10cm as shown in figure. A magnetic field $\vec{B} = (2\hat{i} - 3\hat{j} + \hat{k})$ T is present in the region. (Take the mass of cube 50gm). Match list-I with list-II)

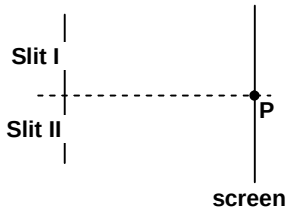


List–I	List–II
(I) magnitude of force on the loop	(P) $\sqrt{150} \times 10^{-2}$ (SI unit)
(II) magnitude of magnetic moment of loop	(Q) $\sqrt{20} \times 10^{-1}$ (SI unit)
(III) magnitude of torque on the loop	(R) $\sqrt{13} \times 10^{2}$ (SI unit)
(IV) magnitude of acceleration of loop	(S) zero

(A) I $\rightarrow$ S, II $\rightarrow$ P, III $\rightarrow$ Q, IV $\rightarrow$ S
(B) I $\rightarrow$ P, II $\rightarrow$ S, III $\rightarrow$ Q, IV $\rightarrow$ S
(C) I $\rightarrow$ S, II $\rightarrow$ P, III $\rightarrow$ S, IV $\rightarrow$ Q
(D) I $\rightarrow$ P, II $\rightarrow$ Q, III $\rightarrow$ R, IV $\rightarrow$ S

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8. A double slit interference pattern is produced on a screen, as shown in the figure, using monochromatic light of wavelength 500 nm. Point P is the location of the central bright fringe, that is produced when light waves arrive in phase without any path difference. A choice of three strips A, B and C of transparent materials with different thicknesses and refractive indices is available, as shown in the table. These are placed over one or both of the slits, singularly or in conjunction, causing the interference pattern to be shifted across the screen from the original pattern. In the list-I, how the strips have been placed, is mentioned whereas in the list–II, order of the fringe at point P on the screen that will be produced due to the placement of the strip(s), is shown. Correctly match both the list.

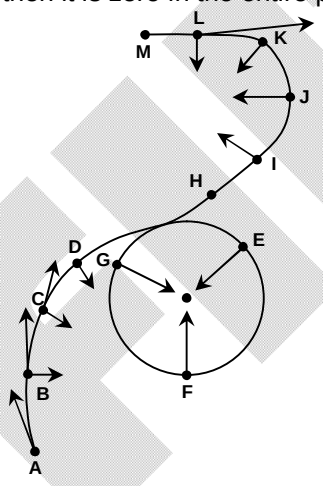


Film	A	B	C
Thickness (in μm)	5	1.5	0.25
Refractive index	1.5	2.5	2

	List–I		List–II
(I)	Only strip B is placed over slit-I	(P)	First bright
(II)	Strip A is placed over slit-I and strip C is placed over slit – II	(Q)	Fourth dark
(III)	Strip A is placed over slit-I and strip B and strip C are placed over slit – II in conjunction.	(R)	Fifth dark
(IV)	Strip A and strip C are placed over slit-I (in conjunction) and strip B is placed over slit – II	(S)	Central bright

(A) I-R; II-R; III-S; IV-P
(B) I-P; II-Q; III-R; IV-S
(C) I-R; II-R; III-P; IV-S
(D) I-P; II-Q; III-S; IV-P

9. Figure shows the path of an ice skater beginning from rest at A to rest at M. The heavy dots along the path indicate the position of the skater every 2s. On the curved path the shown approximate directions and magnitudes for the tangential and centripetal acceleration of several points by arrows.
Note: For the given problem if tangential acceleration at two successive points on curve is zero then it is zero in the entire path between them.

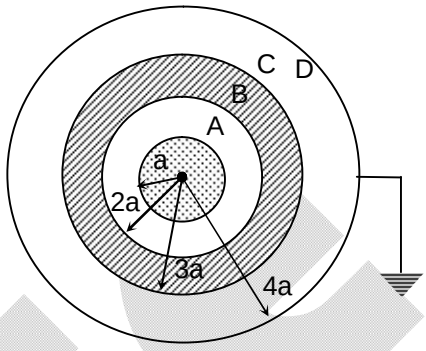


List–I		List–II (Region or Point)	
(I)	Constant speed	(P)	A to C
(II)	No tangential acceleration	(Q)	D to G
(III)	Variable velocity	(R)	I to J
(IV)	Variable magnitude of centripetal acceleration	(S)	E to G
		(T)	No where

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(A) (I) → (Q, R, S); (II) → (Q, R, S); (III) → (P, Q, R, S); (IV) → (P, Q, R)
(B) (I) → (R, S); (II) → (R, S); (III) → (Q, R, S); (IV) → (Q, R)
(C) (I) → (Q, R); (II) → (Q, R); (III) → (P, Q, R); (IV) → (P, Q)
(D) (I) → (Q, S); (II) → (Q, S); (III) → (P, S); (IV) → (P, R)

10. A conducting sphere of radius *a* is placed inside a thin conducting shell of radius 4*a*. Now a dielectric shell of inner radius 2*a* and outer radius 3*a* is placed between sphere and shell concentrically. A is the outer surface of the sphere, B is the inner surface of the dielectric shell, C is the outer surface of dielectric shell and D is the inner surface of conducting shell. The charge $+q$ is given to the outer surface of sphere A.



List –I			List –II	
(I)	Surface A		(P)	Charge is $-q$
(II)	Surface B		(Q)	Charge is $+q$
(III)	Surface C		(R)	Charge is $+q\left(1-\frac{1}{k}\right)$
(IV)	Surface D		(S)	Charge is $-q\left(1-\frac{1}{k}\right)$

(A) I – S; II – Q; III – R; IV – P
(B) I – Q; II – S; III – R; IV – P
(C) I – P; II – R; III – S; IV – Q
(D) I – Q; II – R; III – P; IV – S

Section – B (Maximum Marks: 24)

*This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

11. A small piece of cesium metal (ϕ = 1.9 eV) is kept at a distance of 20 cm from a large metal plate having a charge density of 1.0×10^{-9} C/m^2 on surface facing the cesium piece. A monochromatic light of wavelength 400 nm is incident on the cesium piece. Ratio of maximum to minimum kinetic of the photoelectron reaching the large metal plate is given by k. Then find k.

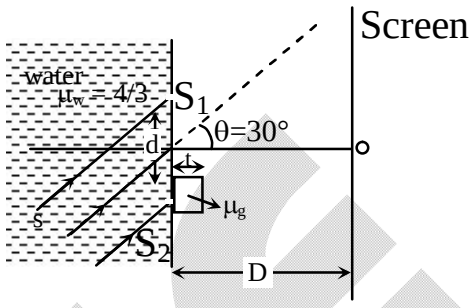
12. Electromagnetic radiation whose electric component varies with time as $E = C_1(C_2 + C_3\cos\omega t)\cos\omega_0 t$, here C_1, C_2 and C_3 are constants, is incident on lithium and librates photoelectrons. If the kinetic energy of most energetic electrons be 0.592×10^{-19} J. Given that $\omega_0 = 3.6 \times 10^{15}$ rad/sec and $\omega = 6 \times 10^{14}$ rad/sec. Find out the work function of lithium in eV (take planks constant $h = 6.6 \times 10^{-34}$ MKS).

13. A railway tank wagon with its 2m diameter and 6m long horizontal cylindrical body, half full of petrol is driven around a curve of radius 100 m, at a speed of 8.33 m/s. The curve runs smoothly into a straight track and the train maintains a constant speed. If value of nf is 2.3. Find the value of 'n'. neglect viscosity and consider petrol as a solid semi cylinder sliding inside the tank. Given: $\tan^{-1}(0.07) \approx 4°$.

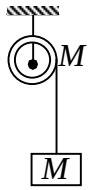
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14. Light of wavelength 1800 Å ejects photoelectrons from a plane of a metal whose work function is 2 eV. If a uniform magnetic field of 5×10^{-5} tesla is applied parallel to plate, what would be the radius of the path (in metre) followed by electrons ejected normally from the plate with maximum energy.

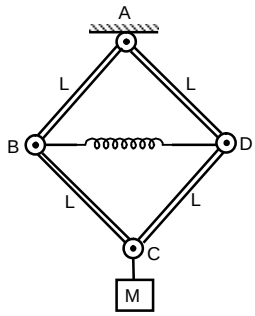
15. In a YDSE experiment, the two slits are covered with a transparent membrane of negligible thickness which allows light to pass through it but does not allow water. A glass slab of thickness t = 0.41 mm and refractive index μ_g = 1.5 is placed in front of one of the slits as shown in the figure. The separation between the slits is d = 0.30 mm. The entire space to the left of the slits is filled with water of refractive index μ_w = 4/3. A coherent light of intensity I and absolute wavelength λ = 5000A° is being incident on the slits making an angle 30° with horizontal. If screen is placed at a distance D = 1m from the slits, then find the distance of central maxima from O (in cm).



16. A mass *M* is supported by a massless and inextensible string wound round a uniform cylinder of mass *M* and diameter 2*R*. With what acceleration in cm/s^2 shall the mass fall when released from rest? (Given *g* = 981 cms^{-2})



17. Four identical mass less rods are connected by hinged joints to form a rhombus of side length $5\sqrt{3}$. Rods can rotate freely about the joints. The joints B and D are connected by a mass less spring of relaxed length $7.5\sqrt{3}$. The system is suspended vertically with a load of mass M attached at C (see figure). In equilibrium the rods form an angle of 30° with the vertical. Find time period of small oscillations of the load.



18. The bulk modulus of rubber is 9.8×10^8 N/m^2. To what depth (in mere) a rubber ball be taken in a lake so that its volume is decreased by 0.1%?

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Chemistry

PART – II

Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

19. Correct statement is/are

(A) $cis-2-butene \xrightarrow[CCl_4]{Br_2}$ Meso compound

(B) Trans-2-butene $\xrightarrow[CCl_4]{Br_2}$ racemic mixture

(C) cis-2-butene $\xrightarrow{Cold\ KMnO_4}$ Meso Compound

(D)  $\xrightarrow[h\nu]{Cl_2}$ 3-monochloro products

20. 1.22 g of C_6H_5COOH is added in two different solvents:

(i) In 100 g $CH_3COCH_3; \Delta T_b = 0.17, K_b = 1.7\,Kg-K/mol$

(ii) In $100\,g\,C_6H_6; \Delta T_b = 0.13, K_b = 2.6\,Kg-K/mol$

Which of the following is/are correct?

(A) Apparent molecular weight of C_6H_5COOH in both the solvents is equal.

(B) Apparent molecular weight of C_6H_5COOH in both the solvents is not equal.

(C) C_6H_5COOH dimerizes in benzene

(D) C_6H_5COOH remains in monomeric form in acetone.

21. White phosphorus (P_4) undergoes redox reaction with:

(A) CS_2

(B) Hot NaOH solution

(C) Chlorine gas

(D) Molten sodium metal

22. Which of the following statements is/are incorrect?

(A) Angular momentum of electron in 3rd Bohr's orbit is $\left(\frac{h}{\pi}\right)$.

(B) 4 moles of H-atoms are present in 40g allene

(C) The value of $(n+l+m)$ for last electron of Na is 3 in its ground state electronic configuration.

(D) The number of electrons in the nucleus of $^{12}_{6}C$ is 6.

23. Which is/are correct statement about following equilibrium

$N_2O_4(g) \rightleftharpoons 2NO_2(g),\ \Delta H = +ve$

(A) Degree of dissociation increases when temperature increases

(B) Vapour density of gaseous mixture decreases when temperature increase

(C) Partial pressure of both N_2O_4 and NO_2 decreases when Helium gas added is added at constant pressure

(D) Partial pressure of N_2O_4 increases when N_2O_4 is added

24. Which of the following complex(s) exhibit structural isomerism:

(A) $\left[Ir(ONO)_3(H_2O)_3\right]$

(B) $\left[Ir(NO_3)(H_2O)_5\right]^{2+}$

(C) $\left[Co(ox)(H_2O)_4\right]Br$

(D) $\left[PdBrCl(NH_3)_4\right]SO_4$

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Section – A (Maximum Marks: 12)

*This section contains **FOUR (04)** Matching List Type Questions. Each question has FOUR statements in **List-I** entries (I), (II), (III) and (IV) and FIVE statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.*

25. Match the List-I and List-II.

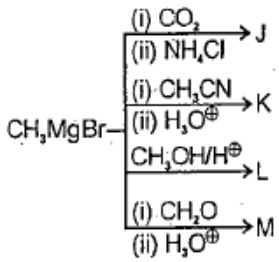
List-I (compounds)		List – II (characteristics)	
(I)	NO_2	(P)	Acidic anhydride
(II)	XeF_2	(Q)	Fluoride acceptor
(III)	OCl_2	(R)	Reduces acidic character of HF
(IV)	BF_3	(S)	Aqueous solution has oxidizing character

(A) I-S, II-Q, III-P, IV–R
(B) I-S, II-R, III-P, IV–Q
(C) I-P, II-Q, III-S, IV–R
(D) I-P, II-R, III–Q, IV–S

26. Complete the following reactions identify the major products and apply a chemical test to distinguish between following pairs:

List- I (Chemical test)		List – II (Pairs)	
(I)	Na-Metal	(P)	J and M
(II)	NaOI	(Q)	J and K
(III)	2, 4-DNP	(R)	K and M
(IV)	$NaHSO_3$	(S)	L and M

Reactions are as follows:



(A) I-R, II-Q, III-P, IV–S
(B) I-S, II-P, III-Q, IV–R
(C) I-Q, II-P, III-R, IV–S
(D) I-P, II-Q, III–S, IV–R

27. Match the List-I and List-II.

List- I (Solution)		List-II (Property)	
(I)	0.1 molal of urea	(P)	$\Delta T_b = 0.25^oC$
(II)	0.1 molar NaCl(aq) at 25ºC	(Q)	$\pi = 4.89$ atm
(III)	0.1 mole PhOH added to 1 kg benzene (*l*)	(R)	$\Delta T_f = 0.2$
(IV)	0.1 mole PhOH $\left(K_a = 10^{-10}\right)$ added to 1 kg water	(S)	Relative lowering = 0.18% in vapour pressure

[Given: $(K_b)_{water} = 0.5\frac{K-Kg}{mol}, (K_f)_{C_6H_8} = \frac{5K-Kg}{mol}, (K_f)_{water} = \frac{2K-Kg}{mol}, (K_b)_{C_6H_8} = \frac{2.5-Kg}{mol.}$

(A) I-Q, II-R, III-P, IV–S
(B) I-Q, II-S, III-S, IV–P
(C) I-R, II-S, III-P, IV–Q
(D) I-R, II-Q, III–P, IV–S

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28. Match the List-I and List-II.

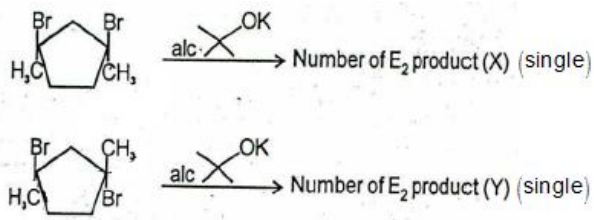
List- I (Ionic Compound)		List – II (Void occupied)	
(I)	NaCl	(P)	100% tetrahedral voids occupied
(II)	ZnS	(Q)	50% Tetrahedral voids occupied
(III)	CsCl	(R)	Octahedral voids occupied
(IV)	CaF_2	(S)	Cubic void occupied

(A) I-R, II-Q, III-S, IV–P
(B) I-Q, II-R, III-S, IV–P
(C) I-R, II-Q, III-P, IV–S
(D) I-Q, II-S, III–P, IV–S

Section – B (Maximum Marks: 24)

*This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

29. How many isomeric aldol condensation products are formed in the given reaction?
Ethyl methyl ketone $\xrightarrow[\text{potash}]{\text{aqueous}}$?

30. 

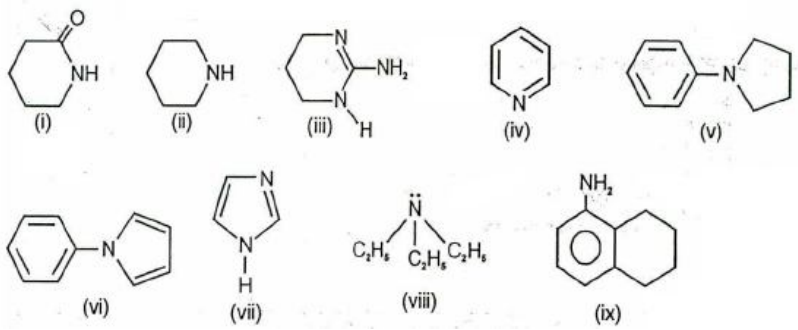
Calculate (X + Y)

31. A compact car gets 20 miles per litre on the highway. Gasoline contains 84% carbon by mass and has a density of 0.80 g/ml. The mass of CO_2 produced (in g) during a 50 mile-trips is

32. Calculate potential (in V) of the cell:
Cu|Mn(s)| $MnCl_2$ (0.001 M), HCl (0.01M)| O_2(0.25 bar) Pt | Cu. Given E^o _ 1.185 V for the Mn^{2+}| Mn couple and 1.229 V for the O_2| H_2O, H^+ couple. (2.303 RT/F = 0.06, log 2 = 0.3)

33. X = number nitrogeneous compounds more basic than 

Aniline



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34. The nucleidic ratio of ${}_{1}^{3}H$ to ${}_{1}^{1}H$ a sample of water is 8.0×10^{-18}:1. Tritium undergoes decay with a half-life period of 12.3 years. How many tritium atoms would 0.9 mg of such a sample contains 36.9 years after the original sample is collected? ($N_A = 6 \times 10^{23}$)

35. The approximate percentage of vacant space in a Silicon cubic cell having crystal structure similar to diamond (round off to nearest integral value)

36. The rate constant for the decomposition of ozone following second-order kinetics is 5×10^{-4} l mol^{-1} s^{-1} at a certain temperature. The rate of reaction (in terms of number of molecules decomposed per ml per minute, when concentration of ozone is 2×10^{-8}M, is ($N_A = 6 \times 10^{23}$)

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Mathematics

PART – III

Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

37. If z be a complex number such that $\left|5z+\frac{1}{z}\right|=3$ and $\arg(z)=\theta$, then $\cos^2\theta$ can be

(A) $\frac{11}{20}$ (B) $\frac{1}{10}$

(C) $\frac{1}{20}$ (D) $\frac{13}{20}$

38. Which of the following statement(s) is(are) true ?

(A) ratio of radius of circumcircle to radius of incircle of an isosceles right angled triangle is $\sqrt{2}+1$

(B) Ratio of radius of circumcircle to radius of incircle of a right angled triangle having angles in AP is $\sqrt{3}+1$

(C) Ratio of radius of circumcircle to radius of incride of in a regular hexagon is $2:\sqrt{3}$

(D) Radius of ex-circle of an equilateral triangle is twice the radius of circum-circle

39. If $2^x + 2^y = 2^{x+y}$, find $\frac{dy}{dx}$

(A) $-\frac{2^y}{2^x}$ (B) $1-2^y$

(C) $\frac{1}{1-2^x}$ (D) $\frac{2^x\left(1-y^2\right)}{2^y\left(1-2^x\right)}$

40. If C_1, $x^2 + y^2 = (3+2\sqrt{2})^2$ be circle. PA and PB are pair of tangents on C_1 where P is any point on the director circle of C_1, then the radius of the smallest circle which touches C_1 externally and also the two tangents PA and PB is

(A) 1 (B) 2

(C) 3 (D) 4

41. If $\theta\in\left[-\frac{\pi}{9},\frac{\pi}{36}\right]$ such that $f(\theta)=\tan\left(\theta+\frac{5\pi}{18}\right)+\cos\left(\theta+\frac{5\pi}{18}\right)-\tan\left(\theta+\frac{7\pi}{9}\right)$ have maximum and minimum value as l_1 and l_2 respectively, then

(A) $l_1=\frac{11}{\sqrt{3}}$ (B) $l_2=\frac{-3+2\sqrt{2}}{2}$

(C) $l_2=\frac{2\sqrt{2}+1}{\sqrt{2}}$ (D) $l_1=\frac{11}{2\sqrt{3}}$

42. Two particles start from the same point (2, -1), one moving 2 units along the line x + y = 1 and the other 5 units along x – 2y = 4. If the particles move towards increasing y, then their new position will be

(A) $(2-\sqrt{2},\sqrt{2}-1)$ (B) $(2\sqrt{5}+2,\sqrt{5}-1)$

(C) $(2+\sqrt{2},\sqrt{2}+1)$ (D) $(2\sqrt{5}-2,\sqrt{5}-1)$

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Section – A (Maximum Marks: 12)

*This section contains **FOUR (04)** Matching List Type Questions. Each question has FOUR statements in **List-I** entries (I), (II), (III) and (IV) and FIVE statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.*

43. Match the list–I & list–II

List–I		List–II	
(I)	Let z, ω, α be complex numbers such that $\|z\| = \|\omega\| = 4$ and $\alpha = \frac{z - \bar{\omega}}{16 + z\bar{\omega}}$, then Re($\alpha$) is equal to	(P)	0
(II)	If $x = p + iq$ is a complex number such that $x^2 = 3 + 4i$ and $x^3 = 2 + 11i$ where $i = \sqrt{-1}$, then (p + q) is equal to	(Q)	3
(III)	Number of complex numbers z satisfying the equation $\bar{z} = iz^2$, where $i = \sqrt{-1}$ is equal to	(R)	4
(IV)	If $z \in C$ satisfies $\|z + 2 - i\| = 5$ then the maximum value of $\frac{\|3z + 9 - 7i\|}{4}$ is equal to	(S)	2
		(T)	5

(A) I $\rightarrow$ Q; II $\rightarrow$R; III $\rightarrow$ T; IV $\rightarrow$ P
(B) I $\rightarrow$ P; II $\rightarrow$Q; III $\rightarrow$ R; IV $\rightarrow$ T
(C) I $\rightarrow$ P; II $\rightarrow$T; III $\rightarrow$ Q; IV $\rightarrow$ R
(D) I $\rightarrow$ P; II $\rightarrow$R; III $\rightarrow$ Q; IV $\rightarrow$ T

44. Match the list–I & list–II

List–I		List–II	
(I)	Poonam flipped a fair coin five times. In the first three flips, the coins came up heads exactly twice and in the last three flips, the coin also came up heads exactly twice. The probability that the third flip was heads, is	(P)	$\frac{3}{4}$
(II)	A box contains 10 transistors of which 2 are defective. Transistors are drawn one by one without replacement unless a non-defective one is chosen. The probability that atmost 3 transistors are drawn, is	(Q)	$\frac{2}{5}$
(III)	A box contains 1 black and 1 white ball. A ball is drawn randomly and replaced in the box with an additional ball of the same colour, then a second ball is drawn randomly from the box containing 3 balls. The probability that the first drawn ball was white given that at least one of the two balls drawn was white, is	(R)	$\frac{4}{5}$
(IV)	Let P(A) = 0.7 and P(B) = 0.3. Let B^c denotes the complement of the event B. Then the smallest value of $P(A \cap B^c)$ is	(S)	$\frac{2}{3}$
		(T)	1

(A) I $\rightarrow$ R; II $\rightarrow$T; III $\rightarrow$ P; IV $\rightarrow$ Q
(B) I $\rightarrow$ R; II $\rightarrow$S; III $\rightarrow$ P; IV $\rightarrow$ Q
(C) I $\rightarrow$ R; II $\rightarrow$P; III $\rightarrow$ Q; IV $\rightarrow$ S
(D) I $\rightarrow$ R; II $\rightarrow$T; III $\rightarrow$ Q; IV $\rightarrow$ S

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45. Match the list–I & list–II.

List–I		List–II	
(I)	Let f(x) is a continuous function. If f(1) = 1 and $\int_0^x t.f(2x-t)dt = \frac{1}{2}\tan^{-1}(x^2)$ then the value of $4\int_1^2 f(x)dx$ is equal to	(P)	0
(II)	Let $f(x) = \prod_{r=1}^{2009}(x-r)$, then the value of the definite integral $\int_1^{2009} f(x)dx$ is equal to	(Q)	1
(III)	If $\int_0^{\ln 10} \frac{e^x\sqrt{e^x-1}}{e^x+8}dx$ has the value of equal to (a – bπ) then $\frac{a}{b}$ equals	(R)	2
(IV)	Suppose g is the inverse function of a differentiable function f and G(x) = $\frac{1}{g(x)}$. If f(4) = 2 and $f'(4) = \frac{1}{16}$ then $(G'(2))^2$ equals	(S)	3
		(T)	4

(A) I → T; II →P; III → Q; IV → S
(B) I → S; II →T; III → Q; IV → R
(C) I → S; II →P; III → T; IV → Q
(D) I → S; II →R; III → T; IV → P

46. Match the list–I & list–II

List–I		List–II	
(I)	Let $f(x)\begin{cases} x^{3/5} & \text{if } x \le 1 \\ -(x-2)^3 & \text{if } x > 1 \end{cases}$ then the number of critical points on the graph of the function is	(P)	5
(II)	Number of real solution of the equation, $\log_2^2 x + (x-1)\log_2 x = 6 - 2x$, is	(Q)	4
(III)	The number of values of c such that the straight line 3x + 4y = c touches the curve $\frac{x^4}{2} = x + y$ is	(R)	3
(IV)	If $f(x) = \int_x^{x^2}(t-1)dt, 1 \le x \le 2$, then global maximum value of f(x) is	(S)	2
		(T)	1

(A) I → R; II →T; III → P; IV → S
(B) I → Q; II →P; III → R; IV → T
(C) I → P; II →Q; III → T; IV → R
(D) I → R; II →S; III → T; IV → Q

Section – B (Maximum Marks: 24)

*This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

47. If a_i, i = 1, 2, 3, 4 be four real numbers of the same sign, then the minimum value of $\sum \frac{a_i}{a_j}$, i, j ∈ {1, 2, 3, 4}, i ≠ j, is

48. Let N = $N = 3^a 2^b 5^c$ where a, b and c are natural numbers and $\{\sqrt{N}\} = 0$. Where {.} denotes fraction part function. Also $f(x) = x^3 - px^2 + qx - 8$ and f(x) = 0 has the roots a, b and c and g(x) is inverse of f(x), then area bounded by y = g(x), x = 8 and y = 0 is :

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49. The value of r for which ${}^{30}C_r.{}^{20}C_0 + {}^{30}C_{r-1}.{}^{20}C_1 + ... + {}^{30}C_0.{}^{20}C_r$ is maximum, is :

50. The equations of the sides AB, BC, CA of a triangle ABC are x + 2y = 0, x + py = q, x – y = 3 respectively. The point P is (2, 3). If P is the orthocentre, then p + q =

51. Consider two matrices $A = \begin{bmatrix} 1 & -2 \\ 3 & 1 \\ -1 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & b-a & -1 \\ a & 1 & 2 \end{bmatrix}$ where a, b $\in$ domain of function $f(x) = \sqrt{-\log_2 \left| \frac{2}{\pi} \sin^{-1}\left(\frac{x+3}{5}\right) \right|}$.

If a, b $\in$ I (set of all integers) then the number of ordered pairs (a, b), is
[Note: Tr.(P) denotes of the trace of square matrix P and adj. (P) denotes the adjoint matrix of square matrix P.]

52. Let function f satisfies
$f(x)f'(-x) = f(-x).f'(x) \forall x \in R, f(0) = 3$
Find f(x) f(–x)+ 2

53. If $\left|\sqrt{3}\cos x - \sin x\right| \geq 2$, for $x \in [0, 4\pi]$, Number of values of x can be n, find n+12 ?

54. There are four boxes A_1, A_2, A_3 and A_4. Box A_i has i cards and on each card a number is printed, the numbers are from 1 to i, A box is selected randomly, the probability of selection of box A_i is $\frac{i}{10}$ and then a card is drawn. Let E_i represents the event that a card with number 'i' is drawn, then
$90\left(P\left(\frac{A_3}{E_2}\right)\right)$ is equal to :

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